



## Introduction

Recent research has shown that chaotic systems are unpredictable in the long run, but their statistical behaviour where trajectories spend time on average is stable and well-defined. This long-run distribution is called the **natural measure**  $\mu^*$ , Chan & Tong (2001) show it can be recovered as **the limit of a noisy system's stationary distribution as noise vanishes**.

This project verifies that claim across **five chaotic systems** (Lorenz, Chen, Lu, logistic map, and tent map), and investigates its robustness under **correlated noise structures** motivated by the error dynamics of **machine learning models** in long-horizon forecasting.

## Chaotic Systems

Five systems are investigated: **the Lorenz, Chen, and Lu systems** (continuous 3D ODEs) and the **logistic map** at  $r = 4$  and **tent map** at  $\mu = 2$  (discrete 1D maps).

The logistic and tent maps are especially useful because their natural measures have **exact closed-form expressions**,  $\mu^*(dx) = \frac{1}{\pi\sqrt{x(1-x)}}dx$ ,  $x \in (0,1)$  and the uniform distribution respectively **allowing direct validation** against ground truth rather than a numerical estimate.

**Lorenz System:**

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= x(\rho - z) - y \\ \dot{z} &= xy - \beta z \\ \sigma &= 10 \quad \rho = 28 \quad \beta = \frac{8}{3}\end{aligned}$$

**Lu System:**

$$\begin{aligned}\dot{x} &= a(y - x) \\ \dot{y} &= cy - xz \\ \dot{z} &= xy - bz \\ a &= 36 \quad b = 3 \quad c = 20\end{aligned}$$

**Chen System:**

$$\begin{aligned}\dot{x} &= a(y - x) \\ \dot{y} &= (c - a)x - xz + cy \\ \dot{z} &= xy - bz \\ a &= 35 \quad b = 3 \quad c = 28\end{aligned}$$

**Logistic Map:**

$$x_{t+1} = rx_t(1 - x_t), \quad r = 4$$

**Tent Map:**

$$x_{t+1} = \mu \min(x_t, 1 - x_t), \quad \mu = 2$$

## Methodology

**Simulating  $\mu^*$ :** A 300,000-step deterministic run from two independent initial condition is used to compute a **finite-sample noise floor** the minimum L1 distance resolvable given finite samples. Any measured distance at or below this floor is **indistinguishable from  $\mu^*$** .

**Noise Generation:** The system is perturbed at each step by additive Gaussian noise,  $X_{t+1} = f(X_t) + \sigma \xi_{t+1}$ . For the 1D maps, a reflection boundary is used to keep the state within  $[0,1]$  without artificially accumulating mass at the boundaries. Each noise level is run for 100,000 steps across 3 independent seeds.

**L1 distance:** The distance between the empirical stationary distribution and  $\mu^*$  is measured by the average absolute bin-by-bin difference across marginal histograms. For the logistic and tent maps, the empirical distribution is compared directly against the exact closed-form natural measure rather than a numerical estimate.

$$L_1(\pi_\sigma, \mu^*) = \int |p_\sigma(x) - p^*(x)| dx$$

References:

- Chan, K.S. & Tong, H. (2001). *Chaos: A Statistical Perspective*. Springer, New York.
- Leonov, G.A. & Kuznetsov, N.V. (2014). Differences and similarities in the analysis of Lorenz, Chen, and Lu systems
- Louw, J. & Ortega, J.P. (2025). Learning the climate of dynamical systems with state-space systems.

## Results of I.I.D Noise Convergence

Across all five systems, the L1 distance between the noisy stationary distribution and  $\mu^*$  **decreases monotonically** as  $\sigma \rightarrow 0$ , confirming the noise-limit characterisation.

The logistic map is validated directly against the **exact arcsine law**, showing the histogram recover the U-shaped density function as noise shrinks. The tent map converges **trivially across all noise levels** to the uniform natural measure is so simple that even large noise cannot visibly distort it. For the three ODE systems, convergence is **confirmed down to the finite-sample noise floor**.

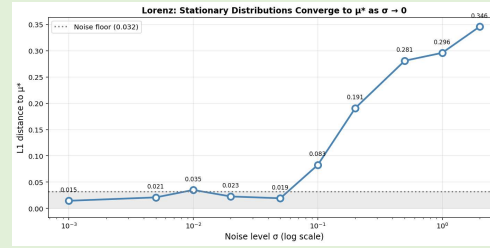


Fig. 1. L1 distance of the Lorenz System compared to its simulated  $\mu^*$  as  $\sigma \rightarrow 0$ .

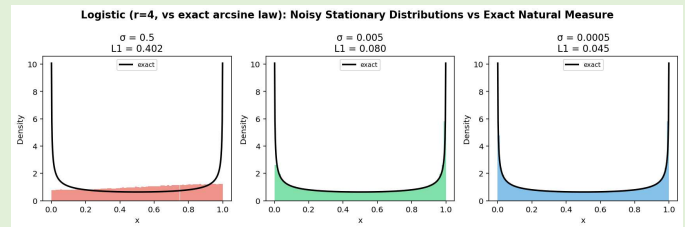


Fig. 2. Histogram of logistic map noisy stationary distributions converging towards the arcsine natural measure as  $\sigma \rightarrow 0$ .

## Correlated Noise Structures

Two correlated noise sources are tested: **AR(1)**,

$$\varepsilon_t = \phi \varepsilon_{t-1} + \sqrt{1 - \phi^2} \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2)$$

where each noise value partially inherits from the previous step (controlled by  $\phi$ ), and **fBm (Fractional Brownian Motion) increments**, where memory decays as a power law (controlled by Hurst exponent  $H$ ). Both model the kind of **temporally correlated errors** produced by ML models in closed-loop forecasting.

## Results of Correlated Noise Convergence

At **fixed  $\sigma$** , stronger correlation pushes the stationary distribution further from  $\mu^*$  but **convergence is never broken**. Furthermore, across **all tested correlation levels**, L1 distance still **decreases toward the noise floor** as  $\sigma \rightarrow 0$  confirming the theorem holds under correlated noise, with stronger correlation requiring smaller  $\sigma$  to take effect.

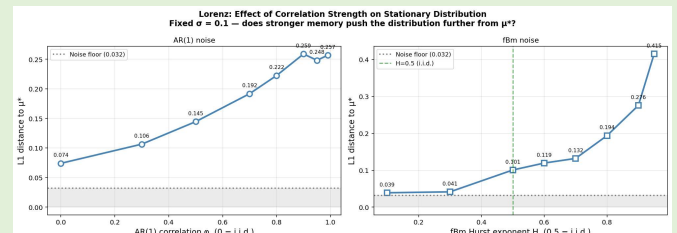


Fig. 3. L1 distance of the Lorenz System to  $\mu^*$  as correlation strength increases at fixed  $\sigma = 0.1$